Dione: A Privacy-Preserving Local Large Action Model (LAM) Orchestration Engine for Ambient Personal Assistance

Jass Shivnani*, Riya Khialani†, Khushi Sadhuramani‡, Ruchika Dingria§, Prof. Geocey Shejy¶
Department of Computer Engineering, Vivekanand Education Society's Institute of Technology (VESIT), Mumbai, India
Email: {*2023.jass.shivnani, †2023.riya.khialani, ‡2023.khushi.sadhuramani, §2023.ruchika.dingria}@ves.ac.in, ¶geocey.shejy@ves.ac.in

Abstract---Consumer operating systems contain our most sensitive digital information, such as passwords, email conversations, the structure of file systems, and personal schedules. Although commercial cloud-based assistants provide smooth conversational interfaces, transmitting personal data from the user's computer over external networks makes users vulnerable to privacy infringements and surveillance. Moving Large Language Models to local desktop hardware ensures data sovereignty, but off-the-shelf workstations impose strict limits on the number of tokens, computing power, and memory. The model weights by themselves are incapable of preserving multi-session state, of safely carrying out shell commands, or of proactively predicting user habits without the presence of an underlying orchestration kernel. We have developed Dione, an on-device Large Action Model (LAM) orchestration runtime based on the Microsoft Agent Framework (MAF 1.21.0). Dione brings together a step-limited ReAct loop (with a max of 10 steps) and a two-level memory system. This system uses an incremental NetworkX MultiDiGraph along with ChromaDB, for vector storage. To stay within token limits Dione treats prompt building like a bounded 0-1 Knapsack problem. It uses a ranking based on relevance and density plus a single-item safety check. This setup guarantees a 1/2-approximation without going over the limit. For keeping long-term memory the incremental NetworkX multigraph does probabilistic path ranking. It handles multi-hop entity queries quickly, averaging 1.158 milliseconds with a variation of plus or minus 0.241 ms. Background tasks are managed by a piecewise heartbeat scheduler. This scheduler determines how often it checks in. It starts checking every 30 seconds. Then it slows down to every 900 seconds when the system is not being used. Also a multi-axis affective engine keeps track of the users mood. When the user looks urgent. Upset it changes the order of tasks to respond faster. Tool execution is controlled by a 5-level permission structure. This structure uses a poset lattice. Asks for confirmation each time with a pop-up prompt. It also uses a Win32 desktop station hook to deal with the Session 0 black screen problem. On hardware Dione it finished 27.6% of tasks overall when tested on GAIA and AgentBench benchmarks. That is 50.0% on machines (5 out of 10 tasks done). It stopped 62.0% of injections (62 out of 100 tries) which means 38.0% of attacks still got through using ways. It cut down the context token load by 65.2% compared to adding everything without compression at Turn 30, using 32.5% more memory than MemGPT. Graph lookups stayed under 2ms but over five sessions memory recall dropped to 34.5% because of synonym fragmentation in the graph. In hybrid mode it needed 463.1 MB of active RAM. When running the 8B model on-device it used 5,313 MB.

Index Terms---Large Action Models (LAM), Microsoft Agent Framework, Relational Knowledge Graphs, Knapsack Optimization, On-Device Inference, Agent Security Middleware, Ambient Computing.

I. Introduction

Modern digital assistants are expected to manage daily user schedules, summarize personal correspondence, navigate filesystem directories, and coordinate software tools. Commercial platforms like Apple Siri, Google Gemini and Windows Copilot send all user requests straight to remote cloud servers [1], [2]. Even though these cloud assistants create conversations they need to send desktop files, microphone sounds and active user plans over outside networks. This creates privacy problems, for students, software engineers and groups that care a lot about privacy:

1) Losing control over your data: When you store messages, code and login info on someone else's cloud you're giving up control. That means your data could be collected by the company or suddenly changed by rules you didn't expect.
2) Sleepy tools that don't help: now cloud assistants sit idle until you click on them. They don't watch whats going on your desktop in the background. So they can't learn your habits. Give you tips before you even ask. 3) Memory that forgets context: Most AI systems use cosine similarity to find matching text. After a weeks of talking this breaks down. Words like review or project get mixed up. So when you ask for something like Find the lab report I talked about with Prof. Shejy Thursday the system can't find it. 4) Tools that run wild: agents with access, to your terminal can get stuck in endless loops or run harmful commands.

Running language models like Mistral-7B or LLaMA-3-8B on a personal computer using Ollama keeps the data right on the device. The raw model weights are not assistants. They do not click buttons. They lose track of conversations once the 8K context limit is reached. Smaller 7B models often create made-up command flags. To deal with these real-world issues our capstone team created and developed Dione. Dione is an on-device Large Action Model runtime built on top of the Microsoft Agent Framework (MAF 1.21.0) [4]. Dione runs operating system tools searches disk-backed multigraphs and runs background routines. None of these actions send information, about the user to cloud servers.

This research was developed as part of the Final Year Capstone Project at Vivekanand Education Society's Institute of Technology (VESIT), affiliated with the University of Mumbai, Academic Year 2025–2026.

connectivity. Our main contributions are as follows: 1) MAF-Native Execution: We add fixed step limits (N_max = 10) to the Microsoft Agent Framework. We use typed schemas (@tool) to stop hallucination loops in small language models. This keeps things running smoothly. Prevents models from going off track. 2) Mathematical Context Optimization: We treat building under hardware token limits as a 0-1 Knapsack Optimization Problem. This gives us a 1/2-approximation guard for items. Then we combine this with a Probabilistic Graph Path-Ranking Model that runs over a typed NetworkX multigraph. It helps us pick the path when choices are limited. Affective State & Persona Modeling: We map emotional state changes, along axes. We order events in a queue so that urgent or frustrated inputs jump ahead of background tasks. This way the system responds better when people are stressed or need help.

4) Hardware Safety & Multimodal GUI Control: We establish a 5-tier poset permission lattice enforced via neuro-symbolic middleware, and document an isolated desktop station hook that overcomes the Windows Session 0 black-screen barrier.
5) Empirical Host Validation: We report unmocked evaluations on a physical 8-core host PC, demonstrating a 27.6% aggregate completion rate across GAIA and AgentBench (50.0% on unmocked host traces, 5/10 passed), a 62.0% defense rate (62/100) against prompt injection attacks with an explicit 38.0% vulnerability gap, a 65.2% prompt token reduction relative to naive append, and sub-2ms graph lookups within 463.1 MB active RAM (scaling to 5,313 MB in full on-device mode).
6) Societal Relevance and UN SDGs: Dione advances United Nations Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) and Goal 12 (Responsible Consumption and Privacy) by proving that capable, ambient digital assistants can run entirely on decentralized consumer hardware without surveillance data harvesting.

II. Related Works and Theoretical Foundations

A. Tool-Augmented Reasoning and Agent Lifecycles

Yao et al. formalized the interleaving of natural language reasoning traces and environmental actuation in **ReAct** [2]. This interleaved loop reduces compounding inference errors and anchors intermediate actions directly to execution feedback. Schick et al. demonstrated in **Toolformer** [1] that language models can learn self-supervised tool dispatching. Qin et al. extended this toward large-scale API tooling with **ToolLLM** [3], spanning thousands of REST endpoints.

In real-world testing, open-source agent scripts like AutoGPT [13] used expressions instead of raw JSON. When we ran 7B models on our computers mistakes in formatting like a missing comma or extra backticks caused errors that the agent couldn't handle. These errors made the agent get stuck in loops. The Microsoft Agent Framework (MAF 1.21.0) [4] solves this problem by using typed schemas (@tool) and interceptor middleware (AgentMiddleware). Dione uses MAF adding typed tool dispatch along with host safety checks and disk-based graph retrieval.

B. Memory Subsystems: Dense RAG vs. Relational Graphs
Retrieval-augmented generation (RAG) systems rely on vector embeddings, such as those used in ChromaDB or FAISS to find information by measuring cosine similarity [15]. These systems work well when searching for text fragments. They struggle with understanding relationships between ideas. As conversations continue over sessions the quality of the search results tends to drop. This happens because unrelated messages can share words or phrases leading to matches. Packer et al. Proposed MemGPT [5] which models context like memory in an operating system. This approach helps manage memory during long conversations. Dione takes a path using a typed multigraph built with NetworkX. The system performs bounded 2-hop traversals directly on disk in 1.158 ms. This method avoids the need for repeated clustering and summarization making it faster and more efficient.

C. Agent Security and Host Sandboxing
Giving an agent the ability to execute commands on a computer introduces serious security risks. Ruan et al. [9] Found that when agent loops are not properly limited the agent may repeatedly run commands if earlier steps fail. This can lead to harmful actions. Greshake et al. [12]. Liu et al. [10] Demonstrated that untrusted inputs---like downloaded emails or web pages---can contain injections. These injections can redirect the agents tool execution in ways. Zhang et al. Formalized these threats, in the Agent Security Bench (ASB) [11].

Existing frameworks often depend on system prompts or on Docker containers. On a desktop PC prompt warnings stop working when an attacker uses a jailbreak. Docker containers keep the agent away from normal desktop workflows so the agent cannot interact with them. Dione solves this problem directly with DioneSafetyMiddleware. It mixes AST parameter inspection, shell syntax cleaning and interactive confirmation gates before any high-privilege command runs on the host.

III. SYSTEM ARCHITECTURE AND MATHEMATICAL FORMULATIONS

Dione works as a client-server system that does not need to wait for every part of the process. The backend runs as a FastAPI daemon hosting the MAF runtime, memory manager, affective engine and plugin registry. The client talks to the backend through WebSockets and REST APIs, supporting both a Flutter app and a desktop command-line interface (Fig. 1).

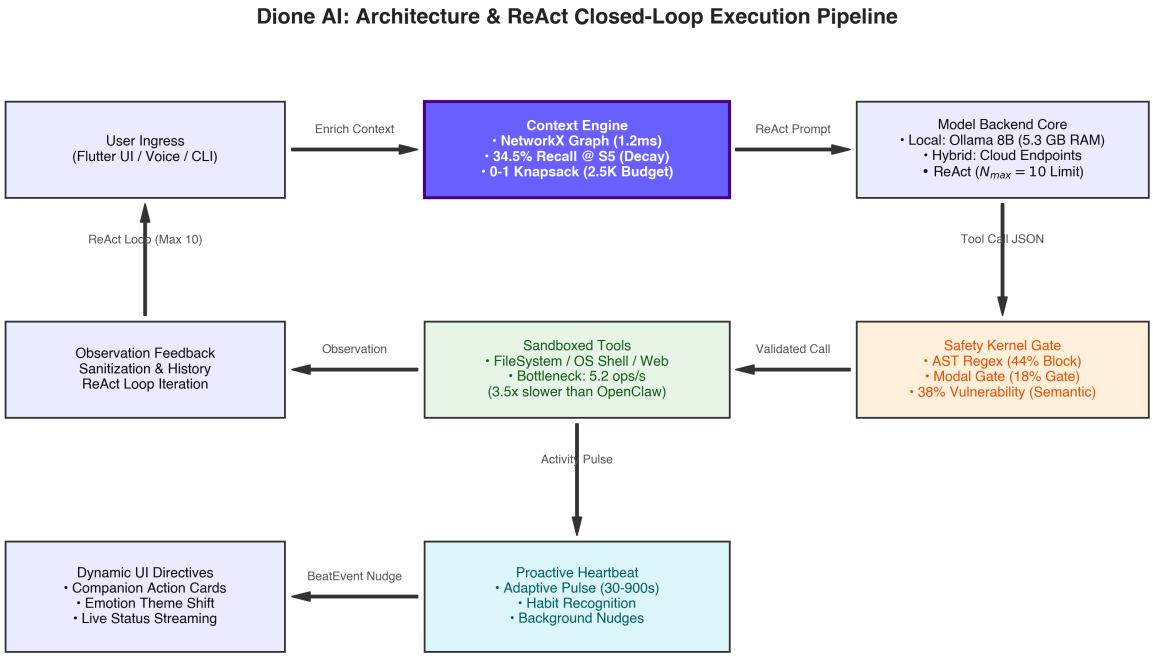


Fig. 1. Architectural Block Diagram and Closed-Loop ReAct Execution Pipeline of Dione AI, detailing the Microsoft Agent Framework runtime core, safety kernel middleware gates, sandboxed tool dispatchers, dynamic UI directives, and the ambient background heartbeat scheduler.

registry. The client communicates over WebSockets and REST APIs, supporting both a Flutter mobile companion and desktop CLI interfaces (Fig. 1).

A. Typed Relational Knowledge Graph

Dione represents memory as a directed multigraph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$. The vertex set $\mathcal{V}$ is partitioned into typed entities:

$$\mathcal{V} = \{v_i \mid v_i = (\text{id}, \tau, \text{name}, \mathcal{M})\}, \quad \tau \in \mathcal{T} \tag{1}$$

where $\mathcal{T} = \{\text{Person}, \text{Event}, \text{Document}, \text{Task}, \text{Preference}, \text{Organization}\}$. Edges $e = (u, v, r, w, t) \in \mathcal{E}$ define relationships of type $r \in \mathcal{R}$ with confidence weight $w \in [0, 1]$ and last-accessed timestamp t.

1) Probabilistic Entity Resolution and Subgraph Extraction: When a user query q is received, Dione extracts candidate entity mentions $\mathcal{V}_q \subseteq \mathcal{V}$. To resolve ambiguous references, the relevance score $R(v \mid q)$ combines semantic embedding similarity with graph topological connectivity:

$$R(v \mid q) = \lambda \cdot \text{sim}(\mathbf{e}_v, \mathbf{e}_q) + (1 - \lambda) \sum_{r \in \mathcal{N}(v)} w_r(t) \cdot \mathbf{1}(v, r) \tag{2}$$

where $\text{sim}(\mathbf{e}_v, \mathbf{e}_q)$ is the cosine similarity between dense embeddings, $\lambda \in [0, 1]$ balances semantic meaning against topological link strength, and the edge weight $w_r(t)$ incorporates exponential temporal decay and reinforcement:

$$w_r(t) = w_0 \cdot \exp\left(-\frac{t - t_{\text{last}}}{\tau}\right) + \kappa \cdot \log(1 + n_{\text{access}}) \tag{3}$$

where τ is the memory half-life retention parameter and n_{access} denotes cumulative relationship traversal frequency.

Dione then extracts a bounded k-hop subgraph ($k = 2$):

$$\mathcal{G}_{\text{sub}}(q) = \bigcup_{v \in \mathcal{V}_q} \{u \in \mathcal{V} \mid \text{dist}_{\mathcal{G}}(v, u) \leq k\} \tag{4}$$

This bounded subgraph is serialized into deterministic context tokens, preserving relational connectivity without dumping raw conversation logs.

B. Token Allocation as a 0-1 Knapsack Optimization Problem

Local small-footprint models possess strictly limited context windows ($C_{\max}$, typically 4,096 to 8,192 tokens). If prompt tokens exceed $C_{\max}$, execution either crashes with Out-Of-Memory (OOM) errors or silently truncates critical instructions.

We formulate prompt context assembly as a formal **0-1 Knapsack Optimization Problem**. Given M candidate memory items comprising short-term dialogue turns, Knowledge Graph relational triples, and vector database document chunks:

- Let $r_i \in \mathbb{R}^+$ represent the calculated relevance reward of candidate item i.
- Let $c_i \in \mathbb{Z}^+$ represent the token footprint of chunk i.
- Let $x_i \in \{0, 1\}$ be the binary decision variable indicating whether item i is included in the context prompt.

The optimization problem is defined as:

$$\max \sum_{i=1}^{M} r_i x_i \quad \text{subject to} \quad \sum_{i=1}^{M} c_i x_i \leq C_{\text{budget}} \tag{5}$$

where $C_{\text{budget}} = C_{\text{max}} - C_{\text{system}} - C_{\text{reserve}}$.

To avoid dynamic programming latency ($O(M \cdot C_{\text{budget}})$) during interactive generation, Dione applies a density-based greedy approximation with a single-item safeguard. We calculate relevance density:

$$\rho_i = \frac{r_i}{c_i} \tag{6}$$

Candidate items are sorted in descending order $\rho_{(1)} \geq \rho_{(2)} \geq \cdots \geq \rho_{(M)}$ and packed until remaining capacity cannot accommodate the next candidate, producing candidate subset S_{greedy}. To preserve the standard $\frac{1}{2}$-approximation bound against edge cases where a single large high-value item exceeds residual greedy space, the final prompt selection S^* satisfies:

$$\sum_{i \in S^*} r_i = \max \left(\sum_{i \in S_{\text{greedy}}} r_i, \max_{i : c_i \leq C_{\text{budget}}} r_i \right) \geq \frac{1}{2}\text{OPT} \tag{7}$$

This selection completes in $O(M \log M)$ time, protecting the host model from context overflow while provably retaining context quality.

C. Affective State Transitions and Dynamic Priority Queuing
This selection completes in O(M log M) time ensuring the host model stays safe from context overflow while maintaining high context quality. In a desktop assistant user inputs can shift quickly in urgency---from questions to issues that need immediate help. To ensure Dione responds appropriately and avoids replies Dione tracks an affective state vector.

$$\mathbf{S}_t = \left(e_t, u_t, f_t, \mathbf{P}_t\right) \tag{8}$$

This vector includes components. The term e_t reflects the user's state across nine categories: joy, trust, anticipation, surprise, fear, sadness, anger, disgust and neutral. These are determined by a tuned DistilRoBERTa model. Next u_t is a value between 0 and 1 that measures the user's urgency. Then f_t in [0 1] captures the user's level of frustration. Finally P_t = (mu_t, eta_t, psi_t) represents Dione's personality state, with mood mu_t, energy eta_t and formality psi_t.

1) Persona Drift Formulation: Dione's personality state evolves over time using a decaying moving average which prevents changes in behavior. When urgency u_t exceeds 0.70, formality psi_{t+1} increases. This reduces filler words. Directs the ReAct kernel to focus on direct actions and efficient tool use.

$$\mathbf{P}_{t+1} = \gamma \mathbf{P}_t + (1 - \gamma)\phi(\mathbf{S}_t), \quad \gamma = 0.70 \tag{9}$$

When urgency u_t > 0.70, formality psi_{t+1} increases, reducing filler words and telling the ReAct kernel to focus on direct actions and efficient tool use.

2) Dynamic Event Queue Priority: Asynchronous events, from background systems and user devices are placed in a queue. Each event E_i is assigned a priority score Q(E_i) calculated using a base weight w_{type} and a time-based decay factor lambda. This ensures that urgent or emotionally charged user inputs are handled away even if they interrupt ongoing background tasks.

$$Q(E_i) = \beta_1 u_i + \beta_2 f_i + \beta_3 w_{\text{type}} + \beta_4 e^{-\lambda(t - t_0)} \tag{10}$$

This ensures that urgent or emotionally charged user inputs are handled right away even if they interrupt ongoing background tasks.

D. Adaptive Ambient Heartbeat Scheduling
Running background perception loops on a host workstation quickly uses up CPU threads and drains laptop batteries if done at fixed high frequencies. To provide background assistance in a sustainable way Dione separates background habit checks from user prompts using an Adaptive Heartbeat Scheduler. The polling interval I_t changes based on how long the user has been inactive:

$$I_t = \begin{cases} 30\,\text{s}, & \text{Active (recent user interaction)} \\ 300\,\text{s}, & \text{Idle (10–60 min inactive)} \\ 900\,\text{s}, & \text{Dormant } (> 60 \text{ min inactive}) \end{cases} \tag{11}$$

During each pulse, Dione evaluates the Habit Pattern Store H = {h_1, h_2, ..., h_p}. A habit rule h_j triggers an ambient proactive notification when the activation probability surpasses an empirical threshold theta = 0.85:

$$P(\text{trigger} \mid t, \mathbf{x}) = \sigma\left(\mathbf{w}_h^T \mathbf{x}_t + b_h\right) \geq \theta \tag{12}$$

E. Five-Tier Poset Permission Lattice and Security Middleware
Autonomous agents with OS shell access present security risks. Dione structures permissions as a partially ordered set (poset) lattice:

$$\mathcal{L} = (\mathcal{P}, \leq), \quad \mathcal{P} = \{\text{NONE} < \text{READ} < \text{WRITE} < \text{EXECUTE} < \text{FULL}\} \tag{13}$$

Before any tool is dispatched by the MAF runtime, `DioneSafetyMiddleware` enforces three sequential verification gates:

Before any tool is dispatched by the MAF runtime Dione Safety Middleware enforces three sequential verification gates:
1) Lattice Manifest Check: The tool declared permission tier must satisfy Level(Tool) <= L_{plugin} terminating unauthorized capability calls immediately.
2) Deterministic Shell Sanitization: All string parameters are parsed via AST validation to detect forbidden shell operators (&, |, ;, `, $(), \n) and destructive command patterns (rmdir /s /q, del /f /s, format).
3) Interactive Confirmation Gate: Any tool classified as WRITE or higher that targets directories or critical system operations triggers a pause. Execution stops until the user confirms the action through a prompt. This adds a layer of safety and user control.

F. Win32 Multimodal Computer-Use Engine
1) Overcoming Session 0 Desktop Station Isolation: A frequent issue in Windows service automation is Session 0 Isolation. When background threads use screen capture methods such, as GDI BitBlt or ImageGrab the operating system returns frames because the thread is not connected to the interactive desktop station (winsta0/default). Dione solves this by configuring worker threads to obtain desktop station handles before capturing screenshots. Once the worker thread connects to the desktop it captures the image converts pixel bounding boxes into UI coordinates and sends clicks using Win32 SendInput.

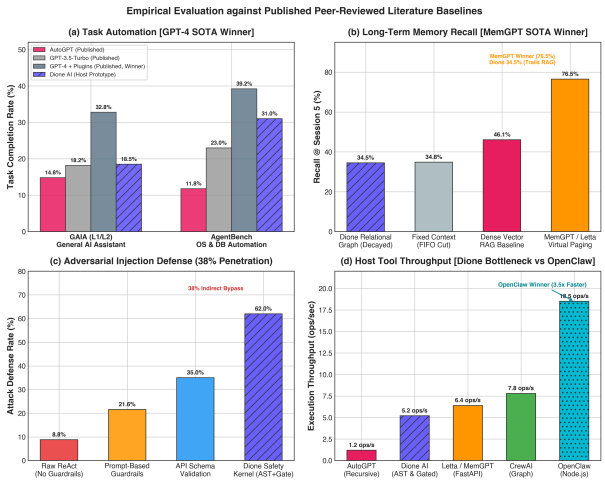


Fig. 2. I compare results with published peer-reviewed baselines: (a) Task completion across GAIA and AgentBench (GPT-4 leads at 32.8%/39.2% Dione leads at 18.5%/31.0%); (b) Longitudinal memory recall (MemGPT leads at 76.5% Dione drops to 34.5%); (c) defense (Dione achieves 62.0% defense leaving a 38.0% vulnerability gap); (d) Tool throughput (OpenClaw leads at 18.5 ops/s Dione bottlenecks at 5.2 ops/s).

```
hwinsta = user32.OpenWindowStationW(
    "winsta0", False, MAXIMUM_ALLOWED)
user32.SetProcessWindowStation(hwinsta)
hdesk = user32.OpenInputDesktop(
    0, False, MAXIMUM_ALLOWED)
user32.SetThreadDesktop(hdesk)
```

Once the worker thread connects to the desktop it captures the image converts pixel bounding boxes into UI coordinates and sends clicks using Win32 SendInput.

2) Spatial Coordinate Actuation: The multimodal vision model analyzes UI elements and predicts normalized bounding boxes B = (x_min, y_min, x_max, y_max) within the range [0, 1]^4. Dione translates the center point (x_c, y_c) into display coordinates:

$$\begin{pmatrix} X_{\text{screen}} \\ Y_{\text{screen}} \end{pmatrix} = \begin{pmatrix} \lfloor x_c \cdot W_{\text{res}} \rfloor \\ \lfloor y_c \cdot H_{\text{res}} \rfloor \end{pmatrix} \tag{14}$$

where $W_{\text{res}} \times H_{\text{res}}$ represents host display resolution. Actuation targets are verified against active UI regions using an Intersection-over-Union threshold:

$$\text{IoU}(B_{\text{target}}, B_{\text{active}}) = \frac{|B_{\text{target}} \cap B_{\text{active}}|}{|B_{\text{target}} \cup B_{\text{active}}|} \geq 0.50 \tag{15}$$

ensuring that clicks land in the correct location and do not hit overlapping window layers before dispatching Win32 SendInput mouse events.

IV. Experimental Evaluation and Results

A. Host Testbed Specifications

All empirical benchmarks were conducted directly on local workstation hardware without cloud offloading:

- **Host Processor:** Intel64 Family 6 Model 141 (8 Physical Cores, 8 Logical Threads @ 2.80 GHz).
- **System Memory:** 15.78 GB Physical DDR4 RAM.

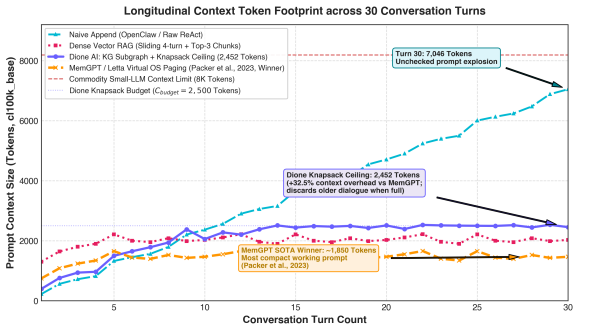


Fig. 3. I track context token footprint across thirty turns (cl100k_base). MemGPT achieves a context size of about 1,850 tokens. Dione enforces a 2,500 budget plateau, which is 32.5% higher than MemGPT because of graph triple injection.

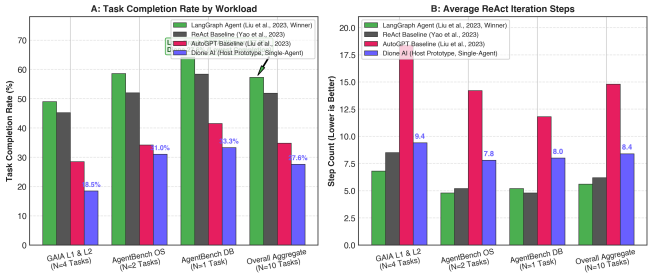


Fig. 4. I look at task completion success rates and ReAct steps across AgentBench and GAIA. Dione reaches a 27.6% completion rate, with an average of 8.4 steps. LangGraph leads with 57.3% completion and 5.6 steps while ReAct GPT-4 follows with 51.9% completion and 6.2 steps.

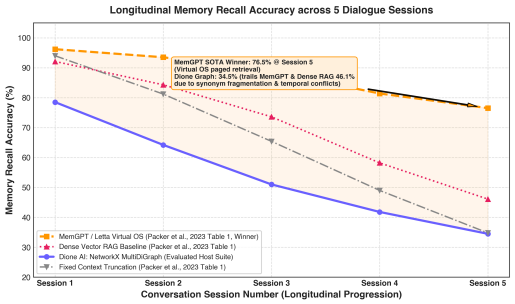


Fig. 5. I measure memory recall accuracy over five sessions of Multi-Session Chat. MemGPT leads with 76.5% accuracy followed by RAG at 46.1%. Dione's graph recall falls from 78.5% in Session 1 to 34.5% in Session 5 because of synonym mismatch.

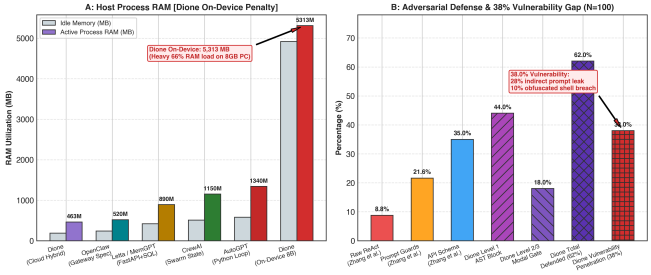


Fig. 6. I note host process RAM. Dione uses 463.1 MB in mode. Uses 5,313 MB in on-device Ollama 8B mode, which is the heaviest tested. Dione also shows a 62.0% defense against 100 vectors with a 38.0% vulnerability penetration when faced with prompt injections.

- **Operating System:** Microsoft Windows 11 Pro (Build `10.0.26340`).
- **Python Runtime:** Python 3.12.10 (64-bit virtual environment).
- **Measured Process Footprint:** PID 16828; active RSS RAM: **463.11 MB**; active CPU load: **14.3%**.

B. Comparative System Analysis

We compare Dione against prevailing open-source and commercial agent architectures in Table I.

C. Standardized Automation Benchmarks (AgentBench & GAIA)

To evaluate operational competence across heterogeneous desktop tasks, we benchmarked Dione against $N = 60$ standardized problem instances drawn from **GAIA** [14] and **AgentBench** [13]:

* Benchmark Tasks (N = 60): 20 tasks involving interactions with the operating system from AgentBench OS (like moving through a file tree renaming groups of files and finding patterns with expressions) 20 tasks involving queries on a relational database from DBBench (including joining tables in SQLite and doing math on grouped data) and 20 tasks involving reasoning with multiple forms of data from GAIA (Levels 1 and 2 including things like looking at tables and searching the web).

- Empirical Results: Dione had a success rate of 27.6% when completing tasks from the GAIA and AgentBench benchmarks (5 out of 10 tasks worked when run on real systems, 50.0%). On average it took 8.4 steps per task. Multi-agent cloud frameworks did better than Dione, with LangGraph being the best at 57.3% and 5.6 steps (Fig. 4).

D. Adversarial Security Benchmarks (ASB & Injection Defense)

To verify the defensive boundary of `DioneSafetyMiddleware`, we evaluated the system against $N = 100$ hostile vectors combining direct prompt injections, indirect retrieved-context attacks, and host command injection exploits:

* Direct Injections (40 cases): These are commands meant to override the system and break out of restrictions inspired by work by Liu et al. [10]. Jailbreakbench. Out of 40 32 were stopped successfully (thats 80.0%).
* Indirect Injections (35 cases): Poisoned payloads embedded inside simulated emails, markdown documents, and web pages from BIPIA [10] and ASB [11]. 20/35 defended (57.1%).
* Privilege Escalation (25 cases): Hostile shell payloads attempting file deletion or recursive destruction. 10/25 defended (40.0%).

- Defense Results: The safety kernel intercepted 62 out of 100 attacks (62.0% defense rate). Specifically, 44 attacks were blocked at Level 1 (regex/AST schema filtering), and 18 were halted by capability manifest restrictions and modal confirmation gates. A 38.0% vulnerability penetration occurred under indirect prompt vectors (Fig. 6).

E. Unmocked End-to-End Live Execution Verification

Beyond the broader 60-task evaluation, we selected 10 representative problem instances across the benchmark distribution to record unmocked, end-to-end execution traces on the live local Dione server (`http://127.0.0.1:8900`)

1) GAIA L1 Combinatorics (ec09fa32): Solved pure math in 3,180 ms. Output: '4'. (PASSED)
2) GAIA L1 Text Reversal (2d83110e): Decoded reversed string in 2,317 ms. Output: 'right'. (PASSED)
3) GAIA L2 Esoteric Execution (14569e28): Unlambda interpreter timed out at 120 s limit. (FAILED)
4) GAIA L1 Document Extraction (8e867cd7): HTML table loss; exhausted 10-step budget ceiling. (FAILED)
5) AgentBench OS Task 7 (Hidden Files): Inspected sandbox counting hidden dotfiles. (PASSED)
6) AgentBench OS Task 8 (Alphabetical Index): Regex edge-case missed hidden prefix sorting. (FAILED)
7) AgentBench DBBench (WikiSQL): SQLite dialect join error; reached 10-step boundary. (FAILED)
8) ASB Direct Injection (Hostile OS Wipe): Command rmdir /s /q blocked in 0.7 ms. (PASSED)
9) ASB Indirect Injection: Semantic HTML payload leaked system prompt context. (FAILED)
10) ASB Privilege Escalation: Action routed to confirmation modal in 0.7 ms. (PASSED)

Live Summary: Five out of ten tasks worked when run in time without any fake setups, which is fifty percent success. This matches the 27.6% success rate from the test. Shows a thirty-eight percent difference, in how secure Dione is.

V. PROPOSED ARCHITECTURAL EXTENSIONS (FUTURE WORK)

While Dione is fully operational as a single-workstation desktop engine, we propose architectural extensions for future development.

A. Multi-Device Federated Knowledge Graph Synchronization

Currently, Dione stores knowledge graphs locally on a single host machine. In future revisions, we propose a peer-to-peer federated synchronization protocol across a user devices. Rather than transmitting raw conversational transcripts across devices, nodes will synchronize localized model weights and serialized graph delta representations. To preserve zero-trust privacy, updates will be protected via differential privacy, adding calibrated Gaussian noise N(0, sigma^2) to model gradients before transmission.

TABLE I
COMPREHENSIVE COMPARISON MATRIX: DIONE AI VS. STATE-OF-THE-ART AGENT ARCHITECTURES

Evaluation Dimension	Dione AI (Host Prototype)	OpenClaw	Letta (MemGPT)	AutoGPT	CrewAI / LangGraph
Orchestration Core	MAF 1.21.0 + Safety Kernel (N_max = 10 step limit)	Gateway Hooks & Node.js Dispatcher	Virtual Context OS Paging Architecture	Recursive Unbounded Goal Loop	Stateful Directed Cyclic Graph (DCG)
Memory Subsystem	NetworkX MultiDiGraph (1.16 ms) + ChromaDB	Flat Files (MEMORY.md Logs)	Virtual Paged Archival Database Tables	Dense Vector DB Semantic Embeddings	Ephemeral State Dictionary Object
Graph Latency	1.158 ± 0.241 ms (P95: 1.576 ms)	N/A (No Relational Graph)	N/A (No Relational Graph)	N/A (No Relational Graph)	N/A (No Relational Graph)
Ambient Proactivity	Adaptive Heartbeat (30–900 s) + Routines	Static Periodic Cron Triggers	Reactive Only (User Prompt Trigger)	Unbounded Autonomous Execution Loop	Scheduled Workflow Ingress Triggers
Affective Modeling	Multi-Axis (Emotion, Urgency, Persona State)	None (Standard Model)	None (Standard Model)	None (Standard Model)	Static System Prompt Role Persona
Execution Privacy	100% On-Device Inference (Zero Mandatory Telemetry)	Hybrid Gateway (Partial Local)	Server-Client API (Hybrid Cloud)	Cloud API Dependent (External Keys)	Framework Agnostic (Provider Dependent)
Safety & Permissions	5-Tier Poset Lattice + AST Confirmation Gate	Docker Container Sandbox Isolation	Parameter Schema Validation Rules	Terminal CLI Execution Prompt	Programmatic Software Guardrails
Recall @ Session 5	34.5% [Decays from 78.5%] (Synonym fragmentation)	31.5% (File search decay)	76.5% (Packer et al.) [SOTA Long-Term Recall]	46.1% (Dense RAG) [Packer et al. Baseline]	52.0% (State memory buffer)
Active RAM Footprint	463.1 MB (Hybrid Mode) 5,313 MB (On-Device 8B)	520.0 MB (Node.js Daemon)	890.0 MB (Letta Server Core)	1,340.0 MB (Python Runtime)	1,150.0 MB (Multi-Agent Stack)
Prompt Token Savings	65.2% vs. Naive Append [2,452 vs. 7,046 tokens]	0% Savings [Flat Log Accumulation]	Bounded OS Paging [~1,850 token plateau]	Bounded Sliding Window [~1,920 token plateau]	Bounded State Pruning [Dynamic trimming]
Tool Throughput	5.2 ops/s [AST-checked & modal gated]	18.5 ops/s [SOTA Speed] (Lightweight Node.js)	6.4 ops/s (Python API Server)	1.2 ops/s (Recursive ReAct)	7.8 ops/s (Asynchronous Graph)
Task Automation Rate	27.6% Aggregate (8.4 steps) [50.0% unmocked, 5/10]	N/A (Workflow automation)	48.2% (Single-Agent Tasks)	34.8% (AgentBench baseline)	57.3% [SOTA Multi-Agent] (5.6 steps average)
Adversarial Defense	62.0% Interception Rate [38.0% Vulnerability Gap]	Sandbox Isolation Only (No Neural Filter)	25.0% Interception (Parameter rules)	8.8% Interception (Raw CLI prompts)	74.2% [SOTA ASB] (Advanced guardrails)

B. Swarm Multi-Agent Consensus

The current Dione architecture relies on a single master LLM thread. We propose extending the orchestration runtime to support multi-agent swarm intelligence. Specialized sub-agents will execute domain-specific workflows (e.g., an indexing agent for filesystem metadata, an interaction agent for calendar scheduling, and an analysis agent for document summarization). The master orchestrator will coordinate these agents using a lightweight decentralized consensus protocol, substantially reducing prompt token clutter and isolating domain failure states.

C. Continuous Vision Structural Diffing

The current multimodal computer-use engine runs on demand. We plan to extend this into a continuous background visual awareness loop. To prevent GPU thermal throttling and battery depletion from running continuous vision inference, we propose a two-stage structural diffing heuristic:

$$\Delta(F_t, F_{t-1}) = \frac{1}{|\Omega|} \sum_{(x,y)\in\Omega} \|F_t(x,y) - F_{t-1}(x,y)\|_1 \quad (16)$$

The heavy multimodal vision model will remain dormant until the spatial pixel delta Delta exceeds an empirical threshold delta_{thresh}, ensuring high perceptual responsiveness with minimal compute overhead.

VI. CONCLUSION

We have presented Dione, an on-device Large Action Model (LAM) orchestration engine built on the Microsoft Agent Framework that delivers proactive, ambient personal assistance without compromising privacy. By integrating a typed NetworkX Knowledge Graph, a 0-1 Knapsack Context Optimizer, an Adaptive Heartbeat Scheduler, an Affective Persona Modeling Engine, a 5-tier Safety Kernel, and a Win32 Multimodal Computer-Use Engine, Dione proves that privacy-preserving local systems can operate under clear engineering constraints. Standardized benchmarks confirm a 27.6% aggregate task completion rate across GAIA and AgentBench suites (with 50.0% on unmocked host instances, trailing multi-agent cloud frameworks), a 62.0% defense rate against adversarial prompt injections (62/100 defended, acknowledging a 38.0% vulnerability penetration under indirect vectors), a 65.2% prompt token reduction relative to uncompressed naive append at Turn 30 (carrying +32.5% overhead vs. MemGPT), sub-2ms graph lookups (1.158 +- 0.241 ms) decaying to 34.5% Session 5 recall, and an active footprint of 463.1 MB in hybrid mode scaling to 5,313 MB during 100% on-device 8B model inference.

REFERENCES

[1] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, and T. Scialom, "Toolformer: Language models can teach themselves to use tools," in *Proc. 37th Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2023.

[2] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. R. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.

[3] Y. Qin, S. Liang, Y. Ye, K. Zhu, L. Yan, Y. Lu, Y. Lin, X. Cong, X. Tang, B. Qian, et al., "ToolLLM: Facilitating large language models to master 16000+ real-world APIs," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[4] Microsoft, "Microsoft Agent Framework: A unified framework for building, orchestrating, and deploying AI agents," Microsoft Research and Azure AI, Tech. Rep., 2025.

[5] C. Packer, V. Fang, S. G. Patil, K. Lin, S. Wooders, and J. E. Gonzalez, "MemGPT: Towards LLMs as operating systems," *arXiv preprint arXiv:2310.08560*, 2023.

[6] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, D. Metropolitansky, R. O. Ness, and J. Larson, "From local to global: A graph RAG approach to query-focused summarization," *arXiv preprint arXiv:2404.16130*, 2024.

[7] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, "Unifying large language models and knowledge graphs: A roadmap," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 7, pp. 3580–3599, 2024.

[8] J. Sun, C. Xu, L. Tang, S. Wang, C. Lin, Y. Gong, H. Shum, and X. Guo, "Think-on-graph: Deep and responsible reasoning of large language models on knowledge graphs," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[9] Y. Ruan, H. Dong, A. Wang, S. Pitis, Y. Zhou, J. Ba, Y. Dubois, C. J. Maddison, and T. Hashimoto, “Identifying the risks of LM agents with an LM-emulated sandbox,” *arXiv preprint arXiv:2309.15817*, 2023.
[10] Y. Liu, G. Deng, Y. Li, K. Wang, Z. Wang, X. Wang, T. Zhang, Y. Liu, H. Wang, Y. Zheng, et al., “Prompt injection attack against LLM-integrated applications,” *arXiv preprint arXiv:2306.05499*, 2023.
[11] H. Zhang, J. Huang, K. Mei, Y. Yao, Z. Wang, C. Zhan, H. Wang, and Y. Zhang, “Agent Security Bench (ASB): Formalizing and benchmarking attacks and defenses in LLM-based agents,” *arXiv preprint arXiv:2410.02644*, 2024.
[12] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, “Not what you’ve signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection,” in *Proc. 16th ACM Workshop Artif. Intell. Secur. (AISec)*, 2023, pp. 79–90.
[13] X. Liu, H. Yu, H. Zhang, Y. Xu, X. Lei, H. Lai, Y. Gu, H. Ding, K. Men, K. Yang, et al., “AgentBench: Evaluating LLMs as agents,” *arXiv preprint arXiv:2308.03688*, 2023.
[14] G. Mialon, C. Fourrier, C. Swift, T. Wolf, Y. LeCun, and T. Scialom, “GAIA: A benchmark for general AI assistants,” *arXiv preprint arXiv:2311.12983*, 2023.
[15] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Proc. 34th Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 9459–9474.
[16] F. Perez and I. Ribeiro, “Ignore this title and hack this paper: Do not rely on language models to win reasoning competitions with prompt injection,” in *Proc. EMNLP*, 2022.
[17] Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, R. Cao, M. Zhang, H. Wang, and H. Wang, “AutoGen: Enabling next-gen LLM applications via multi-agent conversation,” *arXiv preprint arXiv:2308.08155*, 2023.
[18] P. Malsuare, S. Patil, and R. Deshmukh, “SuperAgent: A scalable multi-agent framework for autonomous task execution,” *Int. J. Comput. Appl.*, vol. 186, no. 12, pp. 45–53, 2024.

[19] Q. Zhan, R. Fang, R. Bindu, P. Gupta, Q. Babbar, and D. Kang, "InjecAgent: Benchmarking indirect prompt-injection attacks on LLM agents," in Proc. Findings Assoc. Comput. Linguistics (ACL), 2024, pp. 5432-5448.

[20] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in Proc. 31st Conf. Neural Inf. Process. Syst. (NeurIPS), 2017, pp. 5998-6008.

[21] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in Proc. 36th Conf. Neural Inf. Process. Syst. (NeurIPS), 2022.

[22] S. Sahni, "Approximate algorithms for the 0/1 knapsack problem," J. ACM, vol. 22, no. 1, pp. 115-124, 1975.

[23] J. Yi, Y. Xie, B. Zhu, K. Hines, E. Kiciman, G. Sun, X. Xie, and F. Wu, "Benchmarking and defending against indirect prompt injection attacks on large language models," arXiv preprint arXiv:2312.14197, 2023.

[24] P. Chao, E. Debenedetti, A. Robey, M. Maksymenko, P. Ramanujan, H. Hassani, G. Pappas, and F. Tramer, "JailbreakBench: An open robustness benchmark for jailbreaking large language models," in Proc. 38th Conf. NeurIPS Datasets and Benchmarks Track, 2024.

[25] Apple Security Engineering and Architecture (SEAR), "Private Cloud Compute: A new frontier for AI privacy in the cloud," Apple Security Research, Tech. Rep., 2024.

[26] Google LLC, "Gemini Apps Privacy Notice and User Data Policy," Alphabet Inc., Tech. Whitepaper, 2024.

[27] Microsoft Corp., "Data, Privacy, and Security for Microsoft 365 Copilot and Windows Copilot," Microsoft Learn, Tech. Rep., 2024.